GAME INTEGRITY INVESTIGATION LAB

Synthetic telemetry analysis, detection evaluation, and investigation workflow

Portfolio case study | SQL • Python • SQLite • Behavioral telemetry

Executive Summary

This portfolio project explores how a security analyst can investigate potentially suspicious gameplay behavior using synthetic telemetry. The workflow combines SQL-based player and match analysis, alert review, evidence aggregation, and a comparison between an aggregate-statistics detector and an experimental event-level detector.

The dataset contains 500 synthetic players, 4,650 matches and player-match-stat records, 25,541 gameplay events, and 201 detection alerts. A baseline rule-based detector flagged 8 of 12 players labeled as synthetic cheating cases, with no legitimate players flagged in the reported baseline evaluation. This produced 100% precision, 66.7% recall, and a 0% false-positive rate against the synthetic labels.

An experimental event-level rule caught 10 of the 12 synthetic cheating cases in the comparison performed so far, including three cases missed by the baseline, while missing one case caught by the baseline. This indicates a potential recall improvement on the synthetic cheating subset. The full 500-player confusion matrix for the experimental detector has not yet been calculated, so its precision and false-positive rate remain unverified. The result should not be presented as proof that the experimental detector is better overall.

1. Project Objective and Scope

The objective was to build a hands-on investigation lab that demonstrates a structured approach to reviewing potentially suspicious gameplay behavior. The project is intended as a portfolio demonstration of analytical thinking, SQL investigation, evidence handling, and detection-rule evaluation—not as a production anti-cheat system.

- Summarize player behavior across multiple matches.
- Compare gameplay performance indicators and identify unusual patterns.
- Review repeated alerts and build a focused evidence bundle for a candidate player.
- Evaluate a transparent baseline detector against synthetic ground-truth labels.
- Investigate false negatives and test whether event-level telemetry can recover cases missed by aggregate statistics.

All records are synthetic. The labels represent cases intentionally selected by the data generator for demonstration purposes; they are not independently verified instances of real-world cheating.

2. Dataset and Method

The generator creates 500 players and randomly selects 12 as synthetic cheating cases. Four of those 12 are selected as subtle cases. The remaining 488 players are labeled legitimate; a subset of legitimate players is deliberately generated with higher-skill-looking statistics to make the evaluation less trivial.

The eight non-subtle synthetic cases are generated with elevated target accuracy and headshot-rate distributions, shorter reaction times, and higher movement speeds. The four subtle cases use distributions closer to legitimate behavior. The generator also injects unusually short reaction events and faster movement events for synthetic cheating cases. These patterns are intentional simulation choices, not validated behavioral signatures.

The baseline detector aggregates player-match statistics and alert counts. A player is flagged only when all five conditions are met:

Baseline rule

- At least 5 matches.
- Average accuracy at least 0.56 (56%).
- Average headshot rate at least 0.45 (45%).
- Average median reaction time no greater than 190 ms.
- At least 2 detection alerts.

Because the conditions are joined with AND, failure of any single condition prevents a flag. The rule does not use gameplay_events directly.

3. Investigation Workflow

The SQL investigation was organized into focused questions rather than a single opaque score. The queries examined player-level behavior, ranked versus non-ranked performance, short reaction-time patterns, repeated alerts, candidate evidence, and alert timelines.

- Player behavior summary: aggregated match-level accuracy, headshot rate, reaction time, and movement speed.
- Ranked versus non-ranked comparison: examined performance changes across modes, while recognizing that mode differences alone do not establish misconduct.
- Short reaction-time analysis: examined unusually fast shot reactions as a possible signal requiring context.
- Repeated-alert analysis: identified players with multiple alerts across matches.
- Evidence bundle and timeline: consolidated match statistics and alert history for a candidate case.

The investigation selected player P0428 for a focused evidence review. In match M003990 on 2026-06-28, the recorded match-level statistics included 8 kills, 5 deaths, 61.73% accuracy, 75.91% headshot rate, and a median reaction time of 83.5 ms. An aim-pattern alert associated with that match had a risk score of 0.989. The player's reported average reaction time across matches was approximately 149 ms.

These observations are correlated signals within a synthetic dataset. The alert risk score is not assumed to be a calibrated probability, and the match alone does not prove cheating. The appropriate investigative disposition is to escalate for additional human review, not to declare a definitive violation.

4. Baseline Detector Evaluation

The Python evaluation script compared each detector prediction against the synthetic_ground_truth label and calculated the confusion matrix below.

Baseline confusion matrix

Metric	Count
True positives (TP)	8
False positives (FP)	0
False negatives (FN)	4
True negatives (TN)	488

Derived metrics: precision = 8 / (8 + 0) = 100%; recall = 8 / (8 + 4) = 66.7%; false-positive rate = 0 / (0 + 488) = 0%. These metrics describe performance against the generated labels in this specific toy dataset only.

The four false negatives all met the minimum-match requirement but fell below both the average-accuracy and headshot-rate thresholds. Three also had average reaction times above 190 ms; one had an average reaction time of 188 ms. Three had at least two alerts, while one had only one alert. Because the rule requires all conditions, the accuracy and headshot-rate failures alone were sufficient to prevent flags.

5. False-Negative Analysis and Event-Level Experiment

The generator's code explains why the false negatives require careful interpretation. The label is assigned from membership in the preselected synthetic cheating group, not from whether the player meets the detector's thresholds. The four subtle cases are intentionally generated with target distributions centered around 51% accuracy, 40% headshot rate, and 190 ms reaction time, creating overlap with legitimate behavior.

This does not establish that the labels are wrong. Instead, it shows that the baseline rule is poorly aligned with the subtle group's generated statistics. The event generator also injects short reaction events and faster movement events, but the baseline detector does not use gameplay_events directly.

An exploratory event-level detector was therefore tested. It flags a player if the player has at least 5 matches, at least 2 alerts, at least 10 shot events with reaction times below 120 ms, and at least 10 events with movement speed above 1.25. These thresholds are experimental and are not validated indicators of cheating.

Event-level results on the 12 synthetic cheating cases

Outcome	Baseline	Event-level
Synthetic cheating cases flagged (TP)	8	10
Synthetic cheating cases missed (FN)	4	2
Recall	66.7%	83.3%

The event-level detector recovered three baseline false negatives: P0080, P0277, and P0404. It continued to miss P0040 because that player had only one alert, below the experimental rule's two-alert requirement. It missed P0395, which the baseline caught, because P0395 had 8 short-reaction shots—below the experimental threshold of 10—even though it had 12 high-speed events and 3 alerts.

The event-level rule flagged 10 of the 12 synthetic cheating cases, compared with 8 of 12 for the baseline. This improves recall on the synthetic cheating subset from 66.7% to 83.3%, a gain of 16.7 percentage points. However, only the cheating-labeled rows have been reviewed in this comparison. The experimental detector's false positives among the 488 legitimate players have not yet been counted; therefore, full precision, false-positive rate, and overall comparative performance remain unknown.

6. Findings and Interpretation

- The baseline detector is transparent and easy to explain, but its AND-based rule can miss cases when one or more aggregate metrics fall below fixed thresholds.
- The four subtle synthetic cases expose a known coverage gap: their generated aggregate accuracy and headshot-rate distributions are centered below the baseline thresholds.
- Event-level telemetry can surface patterns hidden by player-level averages. In this experiment it recovered three of the four baseline false negatives, but also missed one baseline true positive.
- A higher recall on cheating-labeled cases does not establish overall improvement. The event-level detector must be evaluated against legitimate players to quantify false positives.
- Ground-truth labels and generator logic are part of the evaluation design. They must be documented and treated as synthetic reference labels, not as independent proof of real-world misconduct.
- Short reaction times and elevated movement speeds are signals to investigate, not proof by themselves. Legitimate explanations, game mechanics, mode, input method, telemetry quality, sample size, and correlated alerts can affect interpretation.

7. Limitations

- All player, match, event, alert, and ground-truth data are synthetic.
- The synthetic labels reflect intentional generator selection, not adjudicated real-world cheating cases.
- The detector thresholds are illustrative and have not been calibrated against real gameplay or reviewed by game-security specialists.
- The event-level detector's confusion matrix across all 500 players is not yet available; its false-positive rate and precision cannot be claimed.
- The event-level thresholds were selected for exploratory comparison. Further threshold testing should use all 500 players and should avoid optimizing solely to catch the known synthetic cases.
- The candidate evidence bundle for P0428 supports further review in the context of this synthetic exercise; it does not support a definitive accusation.

8. Recommended Next Steps

- Run the full 500-player confusion-matrix query for both detectors and calculate precision, recall, and false-positive rate.
- Review the event-level detector's false positives and compare legitimate-player event distributions with those of synthetic cheating cases.
- Test one threshold at a time and record how each change affects true positives, false positives, false negatives, and true negatives.
- Document a final decision rule and retain the baseline as a benchmark for reproducibility.
- Commit the SQL, Python scripts, report, and a concise README with reproduction steps. Include only synthetic data and clearly label all results as illustrative.

Conclusion

This project demonstrates an investigation workflow that connects telemetry analysis, alert review, evidence contextualization, and detector evaluation. The most important result is not that a detector proves cheating; it is that the investigation exposed a measurable gap between the baseline rule and the behavior intentionally injected into subtle synthetic cases. Event-level features recovered three missed cases in the current subset analysis, but full evaluation against legitimate players is still required before claiming an overall improvement.

The project is best presented as a synthetic game-integrity investigation lab that demonstrates SQL analysis, Python-based evaluation, transparent rule logic, and careful interpretation of evidence—not as production anti-cheat experience.